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RESEARCH ARTICLE

Improving Crude Oil Price Prediction Using LLM-Based Hierarchical Semantic Extraction and Heterogeneous Graph Neural Networks

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ABSTRACT Crude oil price forecasting serves energy policy formulation and financial risk management. Existing deep temporal models rely on the assumption of historical pattern continuity and respond insufficiently to exogenous events such as geopolitical conflicts. Mainstream large language model approaches encode news as single vectors, and fine-grained semantic information is compressed and lost in this process. Causal propagation relationships among events and heterogeneous temporal lag characteristics have not yet been incorporated into modeling. To address these limitations, this paper proposes three improvements building upon multimodal graph neural network and Temporal Fusion Transformer architectures. At the semantic extraction level, a four-layer pyramid architecture based on large language models is designed to parse news texts into structured representations covering entity stances, event types, causal chains, and market sentiment. At the graph modeling level, a heterogeneous information graph containing multiple node types and edge types is constructed, introducing causal and temporal edges beyond existing static entity associations to characterize cumulative, propagation, and synergistic effects among events. At the feature fusion level, a semantic-driven market state identification module is designed to replace numerical-statistics-based lagging detection, dynamically adjusting feature weights according to the identified state. Using Brent crude oil price forecasting as a validation case, the proposed method achieves a 17.7% RMSE improvement over the Temporal Fusion Transformer baseline, with improvement reaching 34.7% during event-intensive periods and the coefficient of determination R^2 reaching 0.903.

INDEX TERMS Geopolitical risk analysis, knowledge graph, large language models, oil price forecasting.

I. INTRODUCTION

Crude oil price fluctuations affect the fiscal balance and manufacturing costs of energy-importing nations, and transmit to monetary policy and financial asset valuations through inflation pass-through mechanisms [1], [2]. The price formation of crude oil involves nonlinear coupling among multiple factors including supply-demand fundamentals, geopolitical dynamics, financial speculation, and macroeconomic policy adjustments [1], [2]. Price series exhibit non-stationarity and volatility clustering, with the dominant position of each component dynamically shifting according to market states [3], and these characteristics pose difficulties for predictive modeling.

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Statistical econometric methods such as autoregressive integrated moving average models and generalized autoregressive conditional heteroskedasticity models are constructed based on stationarity and normality assumptions, rendering them unable to adapt to the complex dynamic characteristics of crude oil markets, with prediction accuracy declining during periods of market volatility [3], [4]. In recent years, deep temporal models including long short-term memory networks, gated recurrent units, and Transformers have made progress in crude oil price forecasting tasks by virtue of their nonlinear fitting capabilities and long-range dependency capture abilities [4], [5], achieving improvements over traditional methods across multiple evaluation metrics [6], [7]. However, the common limitation of these methods lies in the fact that their prediction logic relies on the assumption of historical price pattern continuity, with insufficient responsiveness to exogenous shock events such as geopolitical conflicts, extreme weather disasters, and abrupt policy shifts [1], [2]. Such events are among the factors driving large-amplitude crude oil price fluctuations, and this limitation constrains further improvement in the predictive performance of pure time series models.

To compensate for the deficiency of pure time series models in integrating exogenous information, the academic community has begun exploring event-driven prediction paradigms that incorporate unstructured information such as news texts [8]. Early studies utilized the Loughran-McDonald financial sentiment lexicon or methods such as support vector machines to extract news sentiment orientations as auxiliary predictive features [8]. In recent years, large language models such as FinBERT and the GPT series have demonstrated performance superior to traditional methods in financial text analysis tasks [9], [10], providing a new technical pathway for event-driven prediction. Related research indicates that sentiment analysis strategies based on large language models can yield excess returns in stock return prediction [10].

However, several limitations persist in the application of existing large language models to crude oil price prediction. Mainstream methods encode news texts as single dense vectors, and fine-grained semantic information including event-triggering entities, affected objects, and impact directions is compressed and lost during the vectorization process. Models struggle to distinguish event types that are semantically similar yet differ substantially in their impacts [11]. This representational information loss further results in the inability to explicitly model causal propagation relationships among events. The multi-hop causal chains in crude oil markets through which geopolitical conflicts propagate via supply security expectations and hoarding speculation to price formation are simplified into end-to-end mappings [12], leaving models unable to identify critical transmission nodes. Building upon this, the temporal dynamics of event impacts have not yet been incorporated into modeling. Supply-side shocks tend to elicit immediate price responses, while the impacts of demand-side structural changes may persist for several weeks [1], [2]. Existing methods employ unified fusion strategies to handle such heterogeneous latency and decay patterns, making it difficult to capture the differentiated impact trajectories of different event types. The aforementioned limitations progress from event representation through causal modeling to temporal fusion, collectively constraining the performance of large language models in event-driven prediction.

To address these issues, this paper proposes improvements and extensions building upon existing multimodal graph neural network frameworks and the Temporal Fusion Transformer architecture as in Fig. 1. Methods such as MAGNN achieve unified representation of multi-source information by constructing heterogeneous graph structures that fuse historical prices, news texts, and knowledge graph relationships [13]. TFT achieves adaptive fusion of heterogeneous features through variable selection networks and multi-head attention mechanisms [14]. However, existing multimodal graph methods primarily rely on BERT for vectorized encoding of news texts [11], and the constructed graph structures are predominantly based on static entity relationships, making it difficult to characterize the dynamic causal propagation mechanisms among events [12], [13]. The variable selection network of TFT performs feature screening based on numerical statistics and lacks semantic-level understanding of market states [14], [15].

To address these limitations, this paper proposes three improvement measures. At the semantic extraction level, a four-layer pyramid architecture is designed utilizing large language models to sequentially extract entity stance scores, event types and severity levels, inter-event causal graphs, and panic indices with uncertainty indicators from news texts, mapping heterogeneous semantic elements to unified document embedding vectors through weighted fusion functions. At the graph modeling level, a heterogeneous graph is constructed containing four node types comprising news, countries, organizations, and events, along with four relationship types including temporal edges, causal edges, alliance edges, and trade edges. Relational graph convolutional networks are employed for message passing with relation-aware attention mechanisms introduced, enabling the model to dynamically adjust propagation weights for different relationship types according to market states. At the feature fusion level, a large language model-guided market state identification module is designed, determining the current market state through probabilistic fusion of semantic analysis and hidden Markov models, subsequently selecting feature subsets dynamically for input to the Temporal Fusion Transformer for multi-quantile prediction.

The main contributions of this paper are as follows.

1) A four-layer pyramid semantic extraction architecture based on large language models is proposed to address the loss of fine-grained semantic information in flattened encoding.
2) Causal and temporal edges are introduced into the heterogeneous graph, enabling the model to

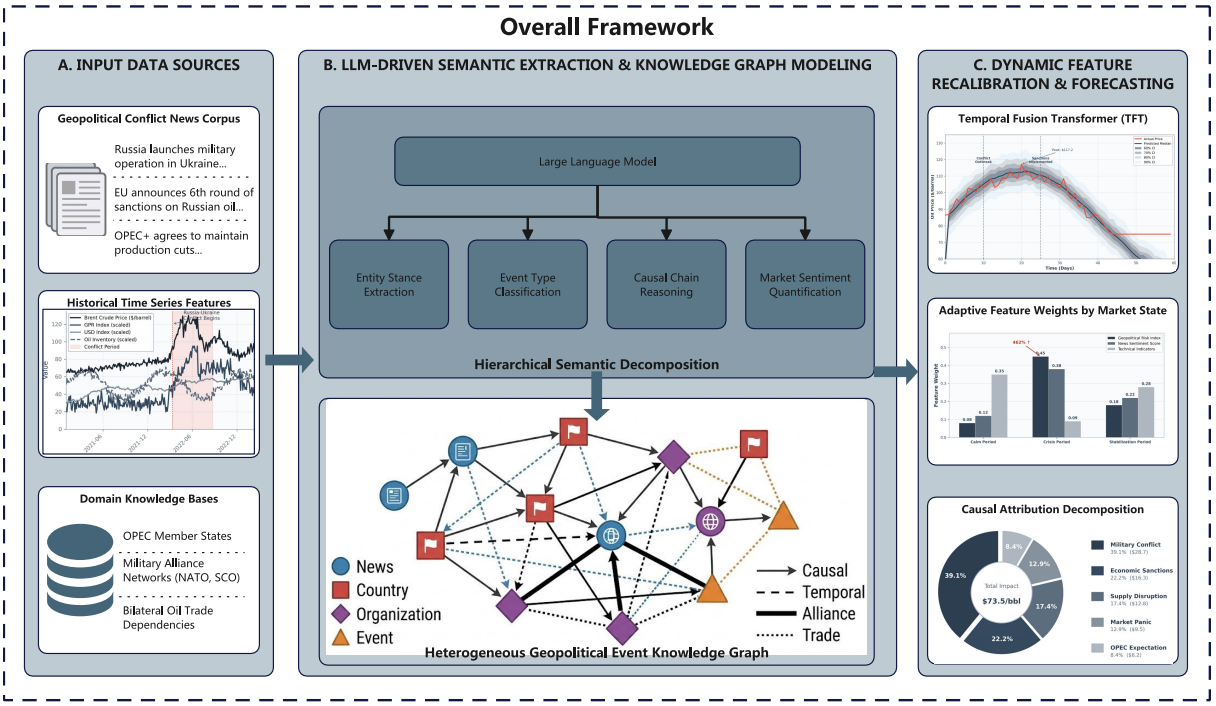


FIGURE 1. Overall architecture of the proposed tripartite forecasting framework integrating LLM-driven hierarchical semantic extraction, heterogeneous geopolitical event knowledge graph modeling, and dynamic feature recalibration for oil price prediction.

characterize dynamic inter-event interaction relationships that existing static graph structures cannot express.

3) A semantic-driven market state identification module is designed to shorten the response latency to state transitions, enabling feature selection to adapt to non-stationary environments.

II. RELATED WORK

A. FINANCIAL TEXT SEMANTIC REPRESENTATION METHODS

Financial texts contain rich market sentiment and event information, and their effective extraction is of significant value for energy price forecasting. Early studies employed the Loughran-McDonald financial sentiment lexicon to annotate news texts, constructing sentiment indices through statistical analysis of positive and negative word frequencies and combining them with traditional machine learning methods such as support vector machines to achieve joint modeling of sentiment features and price series, with related work validating the feasibility of integrating textual information into time series prediction and establishing methodological foundations for subsequent research [16]. The emergence of pre-trained language models has brought paradigmatic innovation to financial text analysis, as BERT acquired deep semantic understanding capabilities through masked language modeling on large-scale corpora [17], [18], domain-adapted models such as FinBERT achieved substantial accuracy improvements over traditional methods in financial sentiment classification tasks [9], and semantic features extracted based on pre-trained models demonstrated stable performance gains across multiple prediction tasks including crude oil prices and stock returns [19], [20], with related achievements having been widely applied and validated in energy economics and financial engineering domains. Existing methods predominantly adopt document-level vectorization strategies when transforming news texts into predictive features, encoding entire reports as single dense vectors before inputting them into downstream models, and while this approach offers computational efficiency advantages, fine-grained semantic information such as the stance orientations of involved entities and the types and severity levels of events is compressed during the encoding process [21], with the powerful capabilities demonstrated by large language models in information extraction and structured output generation in recent years providing new technical possibilities for further enhancing the hierarchical nature of semantic representations [22].

B. MULTIMODAL GRAPH NEURAL NETWORK PREDICTION METHODS

Complex interdependencies exist among entities in financial markets, and effective modeling of these relationships is of significant importance for prediction performance improvement. Knowledge graph technology provides effective tools for structured representation of entity relationships, organizing knowledge scattered across unstructured texts into computable graph structures by representing entities as nodes and relationships as edges [23]. Graph neural networks perform information aggregation on graph structures through

message passing mechanisms [24], graph convolutional networks achieve weighted fusion of neighborhood features using spectral or spatial domain convolutions [25], and graph attention networks adaptively learn importance weights of neighboring nodes through attention mechanisms [26], with both approaches achieving notable success in financial risk assessment, credit scoring, and supply chain analysis tasks. Multimodal heterogeneous graph neural network methods further integrate the aforementioned techniques, wherein the MAGNN framework achieves unified representation of multi-source information by constructing heterogeneous graph structures that fuse historical price series, news text embeddings, and entity knowledge graphs [13], [27], employing metapath-based neighborhood aggregation strategies to capture high-order semantic correlations among different types of nodes [28], [29], and demonstrating favorable prediction performance and interpretability in stock trend prediction and financial event detection tasks, thereby providing a referenceable technical framework for multimodal financial forecasting. Existing graph methods primarily rely on static entity associations when constructing node relationships, such as supply chain relationships among companies or industry affiliations among stocks, while causal propagation mechanisms between events and temporal evolution characteristics of impact effects can be further characterized [30], and introducing causal edges and temporal edges within the heterogeneous graph modeling framework of MAGNN can enhance the representational capacity for complex event network propagation mechanisms.

C. NON-STATIONARY MARKET ADAPTIVE PREDICTION METHODS

Financial markets exhibit significant non-stationary characteristics, with feature importance and optimal prediction strategies dynamically varying according to market states, rendering adaptive modeling methods of considerable value for addressing this challenge. Markov regime-switching models introduce discrete hidden state variables and allow model parameters to switch between different states, enabling characterization of structural changes in prediction mechanisms when markets transition from calm periods to volatile periods [31], and achieving superior performance compared to static models in exchange rate forecasting and volatility modeling. Time-varying parameter methods and online learning strategies continuously update model parameters to adapt to gradual drift in data distributions, further enriching the technical repertoire for adaptive modeling in non-stationary environments [32]. The Temporal Fusion Transformer architecture integrates the aforementioned adaptive concepts with deep learning, performing gated screening of input features through variable selection networks to achieve adaptive allocation of feature importance, capturing sequence dependencies across different temporal scales through interpretable multi-head attention mechanisms, and simultaneously outputting predictions at multiple quantile points through quantile regression to characterize prediction uncertainty [14], [33], with this architecture achieving notable advances in multi-step prediction tasks such as energy prices and electricity load [6], and the feature weights output by its variable selection mechanism providing interpretability support for understanding the predictive contributions of different input variables. The variable selection network of TFT performs screening based on numerical statistical properties of input features, with market state determination relying on cumulative computation of derived indicators such as volatility, resulting in certain response latency for state transitions triggered by sudden geopolitical events [5], [34], and employing semantic analysis techniques to assist market state recognition can capture state transition signals in the early stages of news release thereby improving the timeliness of feature selection [35].

The three categories of methods discussed above involve different technical routes including convolutional neural networks, graph neural networks, and Transformers. The convolution operations of convolutional neural networks are defined on regular grids in Euclidean space, whereas the multiple entity types and relation types involved in event-driven prediction constitute heterogeneous graph structures in non-Euclidean space, on which convolutional neural networks cannot directly operate. Graph neural networks aggregate neighborhood information on arbitrary graph structures through message passing mechanisms, with neighborhoods determined by graph topology rather than fixed windows, making them suitable for modeling irregular relationships among heterogeneous entities. In terms of temporal modeling, convolutional neural networks are constrained by local receptive fields, while Transformers model global dependencies within a single layer through self-attention mechanisms. Based on these considerations, this paper adopts the combination of graph neural networks and Transformers as its technical foundation.

In summary, three categories of methods encompassing financial text semantic representation, multimodal graph neural networks, and adaptive time series prediction have established solid technical foundations for text-enhanced energy price forecasting, with MAGNN representing the current technical frontier in heterogeneous graph modeling and multi-source information fusion while TFT represents the frontier in adaptive feature selection and multi-step distributional prediction. This paper proposes three targeted improvements based on these two mature frameworks, utilizing the structured output capabilities of large language models to enhance the hierarchical nature and granularity of semantic extraction, introducing causal edges and temporal edges within heterogeneous graph structures to enrich the dynamic representation of event relationships, and employing semantic analysis techniques to assist market state recognition to improve the response speed of feature selection to sudden events, aiming to further exploit the predictive potential of existing frameworks rather than fundamentally reconstructing them.

III. METHODS

A. LLM-BASED HIERARCHICAL SEMANTIC EXTRACTION

Event-driven time series forecasting tasks require extracting semantic features relevant to prediction targets from unstructured text. However, existing methods generally encode text into a single dense vector, causing fine-grained information such as event triggers, affected entities, impact directions, and causal propagation to be compressed and lost during vectorization. This representational information loss is particularly pronounced in prediction scenarios with complex semantic structures. When text involves stance differentiation among multiple parties, heterogeneous impacts of different event types, and cross-event causal propagation chains, traditional flattened representations struggle to distinguish semantically similar but impact-divergent event types. Taking oil price prediction as an example, in geopolitical news, production cut announcements from Saudi Arabia and Venezuela have order-of-magnitude differences in their marginal market impacts. Economic sanctions typically involve supply chain adjustment lags of several days, while military conflicts often trigger panic buying that drives sharp price movements within 24 hours. Traditional BERT-based sentiment analysis compresses these differentiated pieces of information into a single scalar, resulting in reduced model responsiveness to complex events as in 2.

Large language models acquire extensive knowledge about various event mechanisms through deep pre-training on massive multi-domain corpora. This knowledge enables models to understand differences in impact lags and propagation paths across event types, and to reason about how events affect prediction targets through multi-hop causal chains. More critically, large language models possess few-shot learning capabilities. Even when facing event types with sparse samples in training data, they can still provide reasonable semantic parsing through contextual reasoning and knowledge transfer. These characteristics make them effective tools for addressing semantic sparsity problems in event-driven prediction.

The advantages of hierarchical representation over flattened representation can be understood from an information-theoretic perspective. Let the text document be D, the prediction target be Y, the traditional flattened representation be $S_{flat} \in \mathbb{R}$, and the hierarchical semantic representation be $S_{hier} \in \mathbb{R}^d$, where dimension d is much greater than 1. For documents containing multi-layered semantic structures, the mutual information of hierarchical representation relative to flattened representation satisfies:

$$I(Y; S_{hier}) \geq I(Y; S_{flat}) + \Delta I \tag{1}$$

where the information gain ΔI originates from three aspects: differentiated entity impacts, event chain reasoning, and semantic context understanding. This information gain can be expressed through conditional entropy decomposition as:

$$\Delta I = H(Y|S_{flat}) - H(Y|S_{entity}, S_{event}, S_{causal}, S_{sentiment}) \tag{2}$$

where $H(Y|S_{flat})$ is the conditional entropy given the flattened representation, and S_{entity}, S_{event}, S_{causal}, $S_{sentiment}$ are the four sub-representations for entities, events, causality, and sentiment, respectively. Since the hierarchical representation contains four sub-representations, its conditional entropy is smaller than that based on a single scalar, thereby providing richer information support for prediction.

Based on the above analysis, this paper designs a four-layer pyramid semantic extraction architecture. The first layer is the entity stance extraction layer, which identifies key entities from text and quantifies their influence tendencies on prediction targets. For each entity e_i, it outputs a stance score $stance_i \in [-1, 1]$ and a relevance weight $relevance_i \in [0, 1]$. The second layer is the event type recognition layer, which constructs domain-relevant event ontologies and determines event type $type_j$ and severity $severity_j \in [0, 1]$. The third layer is the causal chain reasoning layer, which utilizes the multi-hop reasoning capability of large language models to construct causal graphs $G_{causal} = (E, A_{cause})$ between events, with each causal edge associated with a lag parameter τ_{ij}. The fourth layer is the sentiment quantification layer, which extracts sentiment indicators relevant to prediction targets. In the oil price prediction task, the first layer extracts policy stances of oil-producing countries, international organizations, and energy companies. The event ontology in the second layer includes categories such as military actions, economic sanctions, diplomatic negotiations, and supply changes. The fourth layer extracts three market psychological indicators: fear index, uncertainty index, and bullish ratio.

After integrating the outputs of all four layers, the structured semantic representation of document D is defined as a quadruple:

$$S_{LLM}(D) = \left\{ \begin{array}{c} \{e_i : (stance_i, relevance_i)\}, \\ \{ev_j : (type_j, severity_j)\}, \\ G_{causal}, \mathbf{m} \end{array} \right\} \tag{3}$$

where $\mathbf{m}$ is the sentiment indicator vector. To embed this heterogeneous structure into downstream prediction models, this paper designs a weighted fusion function:

$$\begin{aligned} \mathbf{s}_{doc} = \mathbf{W}_e \cdot \phi_{entity} + \mathbf{W}_{ev} \cdot \phi_{event} \\ + \mathbf{W}_g \cdot \phi_{graph}(G_{causal}) + \mathbf{W}_m \cdot \mathbf{m} \end{aligned} \tag{4}$$

where ϕ_{entity}, ϕ_{event}, and ϕ_{graph} are embedding functions for entity sets, event sets, and causal graphs, respectively. Entity embedding is implemented through attention pooling, with weights determined by relevance and attention coefficients $\alpha_i = \text{softmax}(relevance_i)$. Causal graph embedding aggregates topological structure information through graph convolutional networks and generates fixed-dimensional vectors through readout operations. Weight matrices are learned through end-to-end training, allowing each semantic layer's contribution to adaptively adjust according to the prediction task. Hierarchical fusion can capture nonlinear interactions among semantic elements.

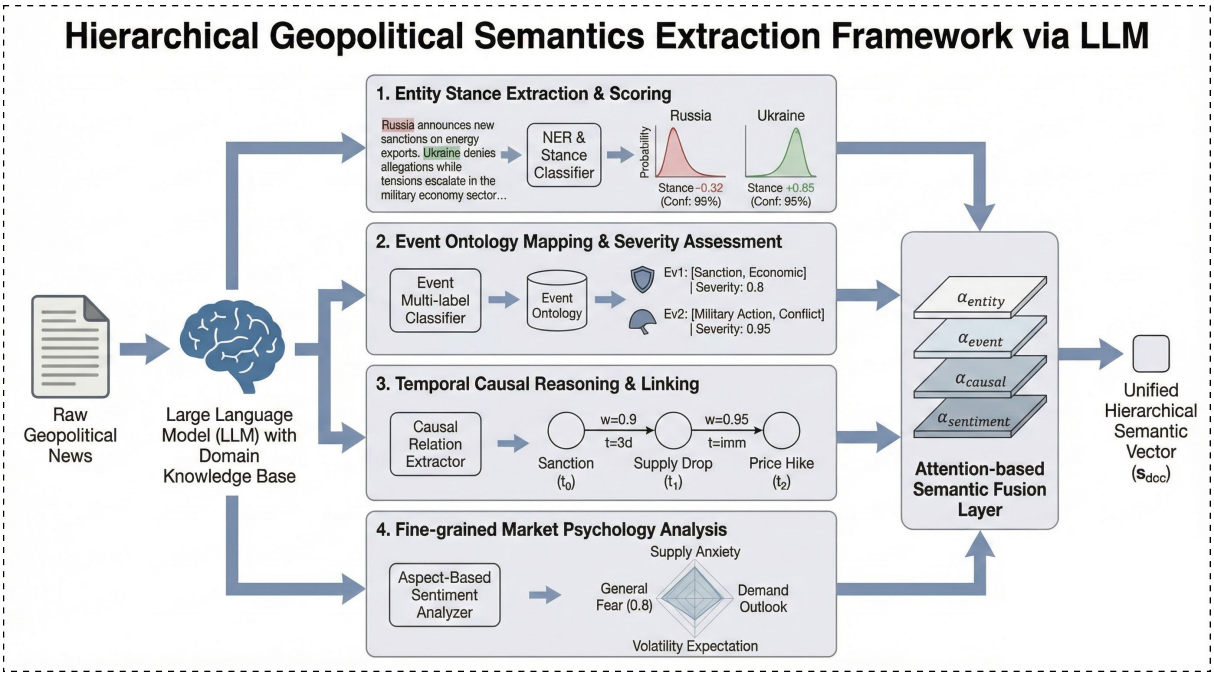


FIGURE 2. LLM-driven hierarchical semantic extraction architecture decomposing geopolitical news into four layers: entity stance extraction, event ontology mapping, temporal causal reasoning, and fine-grained market psychology analysis, unified through attention-based semantic fusion.

The above hierarchical semantic extraction method possesses domain generality, with its core mechanism utilizing pre-trained knowledge of large language models to decompose unstructured text into structured multi-layer semantic representations. This approach can be transferred to other event-driven prediction scenarios, such as extracting virus variant types, control measure intensity, and public sentiment in epidemic spread prediction; extracting supplier status, logistics disruption events, and alternative solution feasibility in supply chain risk warning; and extracting policy signals, corporate dynamics, and market sentiment in financial market prediction. The advantages of hierarchical architecture are particularly significant in scenarios with complex semantic structures. When multiple types of information appear simultaneously and interweave, layered representation can preserve independent information across dimensions and support differentiated utilization by downstream models.

B. HETEROGENEOUS INFORMATION GRAPH CONSTRUCTION AND NEURAL PROPAGATION MODELING

In event-driven prediction tasks, multiple events often exhibit complex interactive relationships rather than isolated effects. This network effect manifests in three forms: cumulative effects where consecutively occurring similar events produce nonlinearly growing impacts; propagation effects where local events spread to broader ranges through dependency networks; and synergistic effects arising from mutual enhancement or offset when multiple actors act simultaneously. Traditional methods simply sum event impacts and struggle to fully capture these structured interaction patterns. Geopolitical conflicts in oil price prediction represent a typical scenario for validating network effects. The Russia-Ukraine conflict case shows that after military actions were triggered, financial sanctions, pipeline suspensions, and reserve adjustments formed cascading reactions within a short period, ultimately driving substantial price increases. Such multi-event network dynamics are far beyond what simple superposition can characterize.

The performance advantages of graph neural networks over flattened methods can be analyzed from a network effect perspective. Flattened models assume event impacts are additive, while nonlinear interactions exist in event networks:

$$\text{Impact}_{total}^{graph} = \sum_{i=1}^{N} \text{Impact}_i + \sum_{i \neq j} \alpha_{ij} \cdot \text{Impact}_i \cdot \text{Impact}_j \quad (5)$$

where the synergy coefficient α_{ij} represents the interaction intensity between events i and j. Graph neural networks implicitly learn these synergistic patterns through multi-layer nonlinear transformations, with performance gains positively correlated with network clustering coefficients and node centrality.

This paper constructs a heterogeneous information graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{R})$ to characterize complex interactive relationships among events. In the general definition, the node set $\mathcal{V}$ contains three types of entities: information source nodes, subject nodes, and event nodes. The edge set $\mathcal{E}$ contains three types of relationships: temporal association edges, causal propagation edges, and structural dependency edges. In the oil price prediction task, information source nodes correspond to news reports, subject nodes correspond to countries and international organizations, and event nodes

correspond to specific events. Structural dependency edges are instantiated as political alliance relationships and trade dependence relationships. The node feature matrix $\mathbf{X}$ is initialized differently according to type: information source nodes are assigned hierarchical semantic embeddings $\mathbf{s}_{doc}$ from the previous module, subject nodes are assigned domain feature vectors, and event nodes are assigned combinations of type encoding and historical impact statistics.

Graph construction integrates automatic extraction and domain knowledge. Temporal edges connect information sources reporting on the same entity within time windows, with weights using temporal decay $w_{ij}^{temp} = \exp(-\lambda \Delta t_{ij})$. Causal edges combine intra-document causal chains with cross-document reasoning, utilizing document-level causal graphs from the previous module and identifying cross-document causal relationships through zero-shot reasoning by large language models. Structural dependency edges are constructed based on external knowledge bases, including oil-producing country alliance membership and bilateral trade dependence in oil price prediction.

Addressing heterogeneous graph characteristics, this paper employs relational graph convolutional networks for message passing. The node update at layer l uses independent transformation matrices for different relation types $r \in \mathbf{R}$:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{W}_0^{(l)}\mathbf{h}_i^{(l)} + \sum_{r\in\mathbf{R}}\sum_{j\in\mathcal{N}_i^r}\frac{1}{c_{i,r}}\mathbf{W}_r^{(l)}\mathbf{h}_j^{(l)}\right) \quad (6)$$

where $\mathcal{N}_i^r$ denotes neighbors connected through relation r, and $c_{i,r}$ is the normalization constant. Relation weights reduce parameter count through basis decomposition $\mathbf{W}_r^{(l)} = \sum_{b=1}^{B} a_{rb}\mathbf{V}_b^{(l)}$. Building upon this, relation-aware attention mechanisms are introduced, enabling the model to dynamically adjust the importance of different relations according to environmental states. Relation-level weights are gated by global context, and node-level attention is computed through relation-specific attention functions. Temporal edges incorporate temporal gating mechanisms, and causal edges follow temporal constraints to ensure causal consistency.

To explicitly model higher-order synergistic effects, a second-order interaction layer is added to capture pairwise interactions among neighbor pairs, which are aggregated to node representations through attention pooling. Final node representations integrate first-order and second-order information, explicitly encoding synergy coefficients. After multi-layer propagation, global graph embeddings are generated through hierarchical pooling:

$$\mathbf{z}_{graph} = [\mathbf{z}_{source} \| \mathbf{z}_{entity} \| \mathbf{z}_{event}] \quad (7)$$

where different node types adopt differentiated pooling strategies. Information source nodes use time-weighted pooling to assign higher weights to recent information, subject nodes use attention pooling, and event nodes use max pooling to extract salient features. This representation encodes cross-document network propagation patterns and structural relationships, complementing the $\mathbf{s}_{doc}$ from the previous module: the latter captures internal semantic hierarchy of individual documents, while the former characterizes documents' structural positions in the macro event network. Residual connections and layer normalization are used to mitigate over-smoothing problems.

The heterogeneous information graph modeling method has broad applicability, with its core lying in explicitly characterizing multi-type relationships among entities through graph structures and capturing interaction patterns through message passing mechanisms. This approach can be applied to financial risk contagion network modeling, social media sentiment propagation analysis, supply chain disruption risk assessment, and other scenarios. Any prediction task where complex dependencies exist among entities and event impacts propagate through networks can draw upon this method. Graph models have advantages in out-of-distribution generalization. When encountering event configurations unseen in training data, they provide additional generalization cues through network structure, making inferences based on structural analogies and demonstrating strong robustness in data-sparse scenarios.

C. SEMANTIC-DRIVEN DYNAMIC FEATURE SELECTION FOR NON-STATIONARY ENVIRONMENTS

Prediction tasks in non-stationary environments commonly exhibit state-switching phenomena, where the predictive value of features differs significantly across states. Static feature selection methods are essentially compromises across all states and are suboptimal in any single state. Let the state space be $\mathcal{S} = \{s_1, s_2, \ldots, s_K\}$, and the state at time t be $s_t \in \mathcal{S}$. Each state has an optimal feature set $\mathcal{F}_s^*$ that minimizes loss in that state. In oil price prediction, market states can be divided into three categories: calm periods, conflict outbreak periods, and stalemate transition periods. Empirical analysis shows that technical indicators dominate during calm periods, geopolitical risk indices and news sentiment become the most important factors during conflict outbreaks, and the predictive value of transportation costs and inventory levels increases during stalemate periods. Traditional static models struggle to capture such dramatic changes in feature importance, leading to significantly amplified prediction errors during state transitions as in Fig 3.

The advantages of dynamic feature selection over static methods can be understood from the perspective of optimization objective decomposition. Decomposing the total loss by state as $\mathcal{L}_{total} = \sum_{s\in\mathcal{S}} p(s)\mathcal{L}_s$, dynamic strategies approximate state-specific optima through state identification $\pi_{dynamic}(s_t) \rightarrow \mathcal{F}_{s_t}^*$, while static strategies selecting fixed feature sets inevitably compromise across different states. Over a sufficiently long time horizon, the average loss of dynamic strategies is smaller than that of static strategies:

$$\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\mathcal{L}(\pi_{dynamic}, s_t) < \lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\mathcal{L}(\pi_{static}, s_t) \quad (8)$$

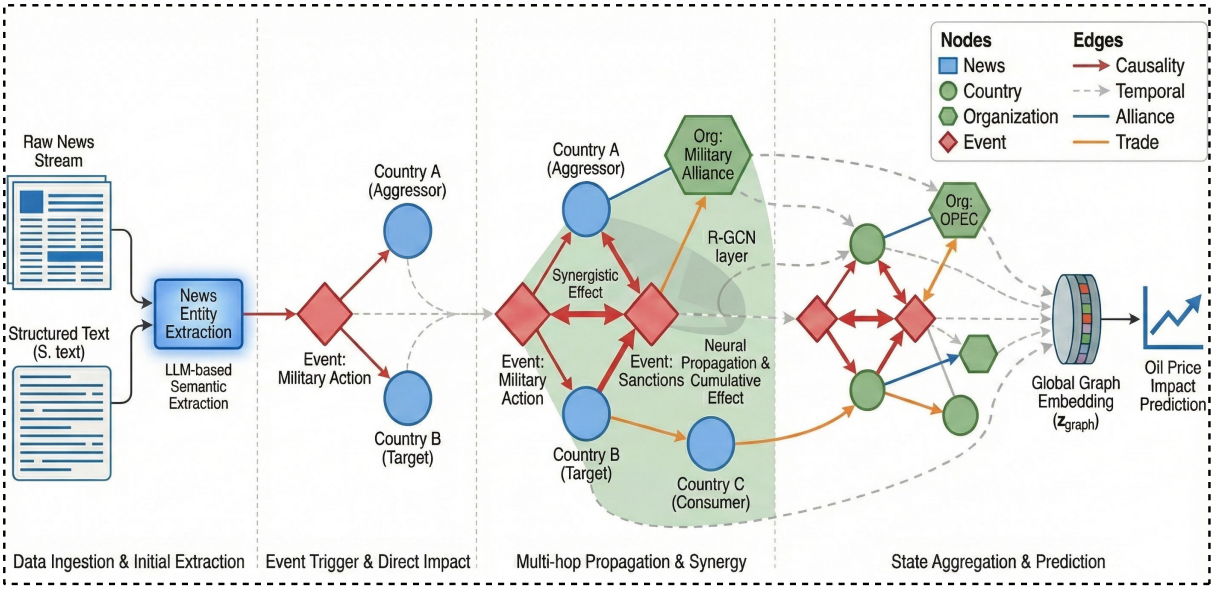


FIGURE 3. Dynamic neural propagation in the heterogeneous geopolitical event knowledge graph, illustrating the temporal evolution from data ingestion through event triggering, multi-hop propagation with synergistic effects, to global graph embedding for oil price impact prediction.

where T is the time horizon, s_t is the state at time t, $\mathcal{L}$ is the loss function, $\pi_{dynamic}$ and π_{static} denote the dynamic and static feature selection strategies, respectively. The above analysis indicates that dynamic selection avoids performance compromises across heterogeneous states by decomposing global optimization into state-specific local optimization.

This paper utilizes large language models to achieve semantic-level state identification, bypassing the lag of traditional numerical methods. State identification combines semantic understanding with temporal modeling. First, large language models perform semantic analysis on the text collection D_t at the current time, outputting probability distributions over states $P_{LLM}(s_t|D_t)$ through few-shot prompting. Simultaneously, hidden Markov models calculate $P_{HMM}(s_t|X_{t-w:t})$ based on historical numerical features and state transition matrices. The two probability distributions are fused through log-odds and then normalized:

$$P(s_t|D_t, X_{t-w:t}) \propto \exp\left(\alpha \cdot \log P_{LLM} + (1-\alpha) \cdot \log P_{HMM}\right) \quad (9)$$

where α is the fusion weight that adjusts according to text information volume. When text is abundant, α is increased; when data is scarce, α is decreased to rely on the temporal smoothing of hidden Markov models. The final state is determined through maximum a posteriori probability.

After determining the environmental state, feature subsets with the highest predictive value are dynamically selected. Feature importance is quantified through gradient-weighted contributions. When data is sufficient, gradient expectations are estimated through multiple forward and backward passes; when data is sparse, prior knowledge from large language models is introduced. The final importance fuses data-driven and knowledge-driven approaches:

$$\alpha_i^{enhanced}(s_t) = \beta \cdot \alpha_i^{gradient}(s_t) + (1-\beta) \cdot \alpha_i^{LLM}(s_t) \quad (10)$$

where β adjusts according to available sample size. The top k features with highest importance are selected to form the dynamic feature set $\mathcal{I}_t$, where k is a hyperparameter balancing expressiveness and overfitting risk.

The dynamically selected feature set is fused with multimodal information to form the input to the Temporal Fusion Transformer:

$$\mathbf{F}_t = [X_t[\mathcal{I}_t] \| \mathbf{s}_{doc}(D_t) \| \mathbf{z}_{graph}(G_t) \| \mathbf{c}_{state}(s_t)] \quad (11)$$

where $X_t[\mathcal{I}_t]$ represents dynamically selected numerical features, $\mathbf{s}_{doc}$ is the hierarchical semantic embedding from the first module, $\mathbf{z}_{graph}$ is the graph embedding from the second module, and $\mathbf{c}_{state}$ is the learnable embedding vector for the state. The Temporal Fusion Transformer encodes on this multimodal input sequence, further adaptively adjusting the contribution weights of each modality through variable selection networks, and finally generates multi-quantile predictions. Multi-quantile outputs provide complete prediction distributions to quantify uncertainty.

Semantic-driven state identification has two advantages over numerical methods. First, it responds quickly. Large language models can identify state transitions within a short time by analyzing key textual content, while numerical statistical methods based on volatility typically require several days of accumulation to achieve statistical significance. Second, it is more robust. Even when encountering event types unseen in training data, large language models can still perform zero-shot generalization based on pre-trained knowledge. Semantic-driven dynamic feature selection methods are applicable to any non-stationary prediction environment with

state switching. Their core advantage lies in utilizing text semantics for rapid state identification. This characteristic also has application value in scenarios such as epidemic phase determination, economic cycle identification, and market sentiment transition detection.

D. JOINT TRAINING STRATEGY AND INTERPRETABILITY ANALYSIS

Jointly training the three improvements of hierarchical semantic extraction, graph neural propagation, and dynamic feature selection with the Temporal Fusion Transformer faces the challenge of coordinating optimization across heterogeneous components. Large language models have massive parameter scales with fixed pre-trained weights, graph neural networks depend on dynamically changing graph structures, feature selection involves discrete optimization, and the Temporal Fusion Transformer requires stable gradient flow. The differences among these components in optimization objectives and learning dynamics make simple end-to-end backpropagation prone to training instability.

This paper adopts a multi-task learning strategy to coordinate different optimization objectives. The overall loss function integrates point prediction loss, interval prediction loss, and regularization loss:

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{point} + \beta \cdot \mathcal{L}_{interval} + \gamma \cdot \mathcal{L}_{reg} \quad (12)$$

where $\mathcal{L}_{point}$ is the point prediction loss using mean squared error, $\mathcal{L}_{interval}$ is the interval prediction loss using weighted sums of quantile losses, $\mathcal{L}_{reg}$ is the regularization loss including weight decay and sparsity constraints, and α, β, γ are the corresponding weight coefficients. These coefficients dynamically adjust according to environmental states, with β increased during periods of high uncertainty.

Joint training adopts a three-stage strategy to mitigate optimization conflicts. The first stage is semantic extraction adaptation, which freezes main parameters of the large language model and only trains low-rank adapter layers. Adapters insert low-rank matrices at each Transformer layer, significantly reducing trainable parameter count. The loss function combines multiple auxiliary tasks including event classification, stance regression, and sentiment prediction. The second stage is graph structure pre-training, which uses graphs constructed from historical data for self-supervised learning. Tasks include node type prediction, link prediction, and subgraph contrastive learning, enabling graph neural networks to obtain good initialization before formal training. The third stage is end-to-end fine-tuning, which unfreezes all modules for joint optimization. At this point, each component already possesses basic functionality, and fine-tuning mainly adjusts inter-module interfaces and weight allocation. Hierarchical learning rates are adopted to reflect parameter scale differences across components, and gradient clipping and accumulation strategies are used to stabilize the training process.

This paper analyzes model interpretability from two perspectives: internal mechanisms and external attribution. Internal interpretability utilizes the variable selection network and temporal attention mechanism of the Temporal Fusion Transformer. Variable selection weights reflect marginal contributions of features, and temporal attention weights reveal the influence of historical moments on predictions. External interpretability performs feature attribution through Shapley values. Shapley values satisfy axiomatic properties such as efficiency, symmetry, and monotonicity, enabling fair quantification of each feature's contribution to predictions. This paper computes rank correlation coefficients between variable selection weights and Shapley values to verify consistency between the two interpretability methods.

Counterfactual analysis is used to deepen understanding of prediction mechanisms, answering how prediction results would change if certain types of information were removed. Counterfactual prediction is implemented through graph structure intervention. After removing specific types of nodes or edges from the graph neural network, graph embeddings are recomputed and predictions are generated. Causal effects are quantified as the difference between actual predictions and counterfactual predictions. Multi-factor contribution decomposition attributes overall prediction changes to different information sources. Each factor's contribution is obtained by progressively removing corresponding features and computing prediction changes. For cases with strong interactions, Shapley interaction indices are used to quantify pairwise synergistic effects.

Interpretability analysis relies on causal inference assumptions such as unconfoundedness and positivity. Unconfoundedness is approximately satisfied by integrating multimodal information. Positivity is difficult to satisfy when certain event configurations have not historically occurred. In such cases, analogical reasoning based on similar historical cases can be performed through the knowledge base of large language models. The above interpretability analysis methods are applicable to various multimodal prediction tasks, helping users understand prediction bases and evaluate contributions from different information sources. In oil price prediction, these analyses provide quantitative evidence for understanding market price formation mechanisms.

IV. EXPERIMENTS AND RESULTS

A. EXPERIMENTAL SETUP

1) DATASET CONSTRUCTION

This study uses daily data from January 2021 to July 2024 for experimental validation. Oil price data adopts Brent crude oil spot prices provided by the U.S. Energy Information Administration, covering 870 trading days spanning before and after the Russia-Ukraine conflict. News data was collected through web scraping from Reuters and Oilprice.com, filtered using keywords including Russia, Ukraine, oil, sanctions, and energy, yielding approximately 4,500 geopolitically relevant reports. Auxiliary economic variables include the Geopolitical Risk Index, Economic Policy Uncertainty Index, U.S. Dollar Index, S&P 500 Index, crude oil inventory, and OPEC production data, all sourced from publicly available databases.

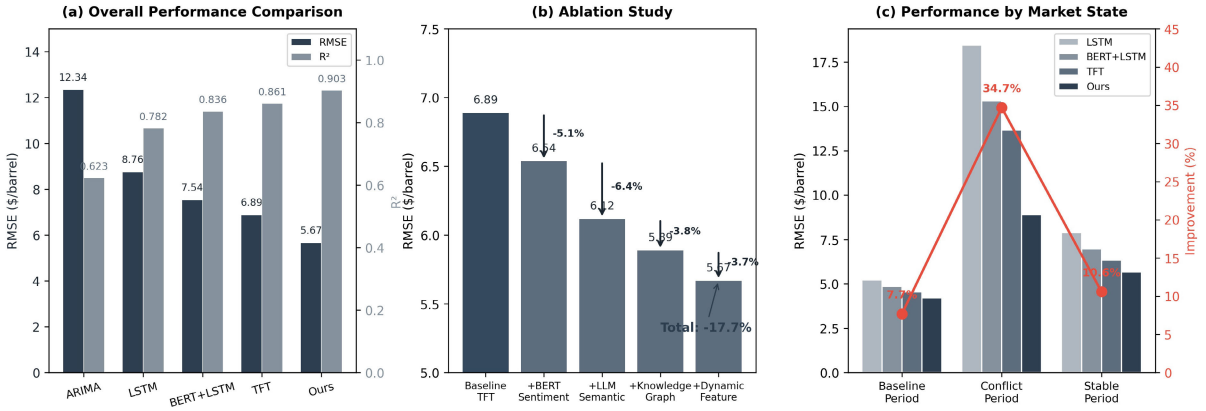


FIGURE 4. Experimental results: (a) overall performance comparison showing RMSE and R^2 across all models, (b) ablation study demonstrating incremental contributions of each module with a total improvement of 17.7%, and (c) performance across different market states with 34.7% improvement during the conflict period.

Data was divided chronologically into three non-overlapping subsets. The training set covers January 2021 to December 2022, comprising 70% of the total samples, including both the pre-conflict baseline period and the conflict outbreak period, used for model parameter learning. The validation set spans January 2023 to June 2023, comprising 15%, used for hyperparameter tuning and early stopping decisions. The test set covers July 2023 to July 2024, comprising 15%, strictly reserved for final performance evaluation to ensure unbiased results.

2) EVALUATION METRICS

Prediction performance is quantified through four complementary metrics. Root Mean Square Error (RMSE) measures the overall deviation between predicted and actual values, being more sensitive to large errors and suitable for capturing extreme volatility. Mean Absolute Error (MAE) provides a linear measure of error, more robust to outliers. Mean Absolute Percentage Error (MAPE) expresses relative error in percentage form, facilitating cross-period comparisons. The coefficient of determination R^2 measures the model's ability to explain price variance, with values closer to 1 indicating better fit. All metrics are calculated on the test set without any post-processing or adjustments.

3) BASELINE MODEL SELECTION

To comprehensively evaluate the effectiveness of the proposed method, six representative baseline models were selected. ARIMA represents traditional econometric methods, modeling through the autocorrelation structure of time series. LSTM, as a typical representative of recurrent neural networks, captures long-range dependencies in sequential data. Informer [5] reduces the computational complexity of self-attention through the ProbSparse mechanism and employs a generative decoder for efficient long-sequence prediction. Autoformer [6] introduces series decomposition and autocorrelation mechanisms, separating trend and seasonal components for independent modeling. The BERT combined with sentiment analysis and LSTM hybrid model represents text-enhanced prediction methods, where BERT extracts flattened sentiment scalars from news as additional inputs to LSTM. The Temporal Fusion Transformer without news information uses only numerical features, achieving adaptive feature fusion through variable selection networks and multi-head attention mechanisms. Informer and Autoformer use the same numerical feature inputs as TFT without news information. These six baseline models cover the technical evolution path from traditional econometrics to recurrent networks, Transformer variants, and multimodal fusion.

4) IMPLEMENTATION DETAILS

All deep learning models were trained on an NVIDIA RTX 3090 GPU using the PyTorch 2.0 framework and Python 3.9 environment. The large language model uses the GPT-4 API for hierarchical semantic extraction, with processing time of approximately 1.5 seconds per news article. The relational graph convolutional network is configured with 3 layers, hidden dimension of 256, using basis decomposition to reduce parameter count. Model training employs the Adam optimizer with learning rate set to 1e-4, batch size of 32, maximum 50 training epochs, and early stopping strategy to prevent overfitting. Hyperparameters were determined through grid search on the validation set, and all experimental results are averages of 5 independent runs with standard deviation controlled within 5%.

B. OVERALL PERFORMANCE VALIDATION

Table 1 presents the prediction performance comparison of all models on the test set. Traditional ARIMA achieves an RMSE of \$12.34 with R^2 of only 0.623, indicating that linear models experience decreased prediction accuracy in volatile market environments. Deep learning LSTM reduces error to \$8.76, but R^2 remains at only 0.782, leaving 21.8% of price variance unexplained. Informer reduces computational complexity through the ProbSparse attention mechanism, achieving an RMSE of \$7.82 and R^2 of 0.812, improving

over LSTM but with limited adaptability to non-stationary sequences. Autoformer introduces series decomposition and autocorrelation mechanisms, further reducing RMSE to \$7.23 with R^2 of 0.843, as its decomposition architecture outperforms standard attention mechanisms in capturing trend and seasonal components. The BERT-Sent plus LSTM model incorporating news sentiment reduces error to \$7.54, validating the predictive value of textual information, but flattened sentiment loses fine-grained information such as entity stances, event types, and causal chains.

TABLE 1. Overall prediction performance comparison on test set.

Model	RMSE	MAE	MAPE	R^2
ARIMA	12.34	9.87	11.43%	0.623
LSTM	8.76	6.89	8.12%	0.782
Informer	7.82	6.14	7.23%	0.812
Autoformer	7.23	5.67	6.71%	0.843
BERT-Sent+LSTM	7.54	5.92	6.98%	0.836
TFT without news	6.89	5.34	6.27%	0.861
Ours	**5.67**	**4.38**	**5.13%**	**0.903**

Note: All error metrics are in USD per barrel; R^2 is dimensionless; optimal values are marked in bold. Informer and Autoformer use the same numerical feature inputs as TFT without news information.

The Temporal Fusion Transformer achieves an error of \$6.89 with R^2 of 0.861, but performance decreases during event-intensive periods. Notably, the R^2 of Autoformer at 0.843 is slightly higher than that of BERT-Sent+LSTM at 0.836, indicating that architectural improvements and the introduction of flattened textual information yield gains of comparable magnitude, while the hierarchical semantic extraction and graph modeling in this paper incorporate the advantages of both aspects simultaneously. The proposed method reduces RMSE to \$5.67, representing a 17.7% improvement over the TFT baseline, a 21.6% improvement over Autoformer, and a 27.5% improvement over Informer, with R^2 reaching 0.903. MAE is \$4.38 with an error rate of approximately 5.2%, below the industry-acceptable threshold of 6%.

To verify the statistical significance of performance differences, Diebold-Mariano tests were conducted for all model pairs. The DM statistic between the proposed method and the TFT baseline is -3.42 with $p < 0.001$, -4.17 with $p < 0.001$ against BERT-Sent+LSTM, -3.89 with $p < 0.001$ against Autoformer, -4.53 with $p < 0.001$ against Informer, and -5.83 with $p < 0.001$ against LSTM. All differences among ablation variants are statistically significant at the 0.05 level. All experimental results are averaged over 5 independent runs with standard deviations kept within 5%.

Performance improvements stem from the synergistic effects of the three improvement measures. Hierarchical semantic extraction captures entity stances, event severity, and causal chain reasoning, providing richer information support compared to flattened sentiment. Heterogeneous graph modeling characterizes event propagation effects in networks, identifying multi-event accumulation and synergistic amplification patterns. Dynamic feature selection, through semantic-driven state identification, increases relevant risk indicator weights from 8% to 45% during event-intensive periods.

C. ABLATION STUDY AND MODULE CONTRIBUTIONS

Table 2 presents systematic ablation experiment results. Starting from the baseline TFT, four modules were progressively added: BERT flattened sentiment, LLM hierarchical semantics, heterogeneous graph modeling, and dynamic feature selection.

BERT flattened sentiment provides a 5.1% improvement, validating the auxiliary role of news text for prediction. Replacing it with LLM hierarchical semantics yields a single-step improvement of 6.4%, the largest contribution among all modules. Heterogeneous graph modeling contributes 3.8%, with graph structure capturing cumulative effects, propagation effects, and synergistic effects among events. Dynamic feature selection contributes 3.7%, analyzing daily text to determine environmental states and adjust feature weights. The total improvement is 17.7% versus a theoretical cumulative of 19.0%, with the difference attributable to slight functional overlap among modules, but each module's marginal contribution is strictly positive.

To further validate the contribution of each semantic layer within the four-layer pyramid architecture, Table 3 presents the ablation results of removing individual layers. The dynamic feature weight patterns across different market states are illustrated in Fig. 5.

Removing the causal chain layer leads to a 5.6% increase in RMSE, the largest impact among all four layers. Cross-event causal reasoning provides structural information that cannot be substituted by other layers. Removing the entity stance layer results in a 4.2% RMSE increase, the event type layer a 3.0% increase, and the sentiment indicator layer a 2.1% increase. When only the sentiment indicator layer is retained, RMSE rises to 6.41, close to the 6.54 of BERT flattened sentiment. This result indicates that a single sentiment dimension has similar information content to flattened methods, and that the advantage of the hierarchical architecture stems from the joint encoding of multi-dimensional information.

D. PERIOD-SPECIFIC PERFORMANCE ANALYSIS

Table 4 presents prediction performance comparisons across different market states. The baseline period spans January 2021 to February 2022, when markets were in post-pandemic recovery with relatively moderate price fluctuations. The event-intensive period covers March to June 2022, encompassing the geopolitical conflict outbreak and subsequent chain reactions, with volatility rising from 8% to 35%. The stable period corresponds to the July 2023 to July 2024 test set, when markets adapted to the new landscape with volatility falling back to 18%.

During the baseline period, all models perform relatively similarly with errors ranging from \$4.21 to \$5.23. In calm

TABLE 2. Ablation study results and module contribution analysis.

Model Variant	RMSE	MAE	R^2	Improvement
Baseline TFT	6.89	5.34	0.861	-
+BERT flattened sentiment	6.54	5.08	0.876	5.1%
+LLM hierarchical semantics	6.12	4.76	0.891	6.4%
+Heterogeneous graph modeling	5.89	4.56	0.897	3.8%
+Dynamic feature selection	**5.67**	**4.38**	**0.903**	3.7%

Note: Improvement is calculated as the percentage reduction in RMSE relative to the previous row.

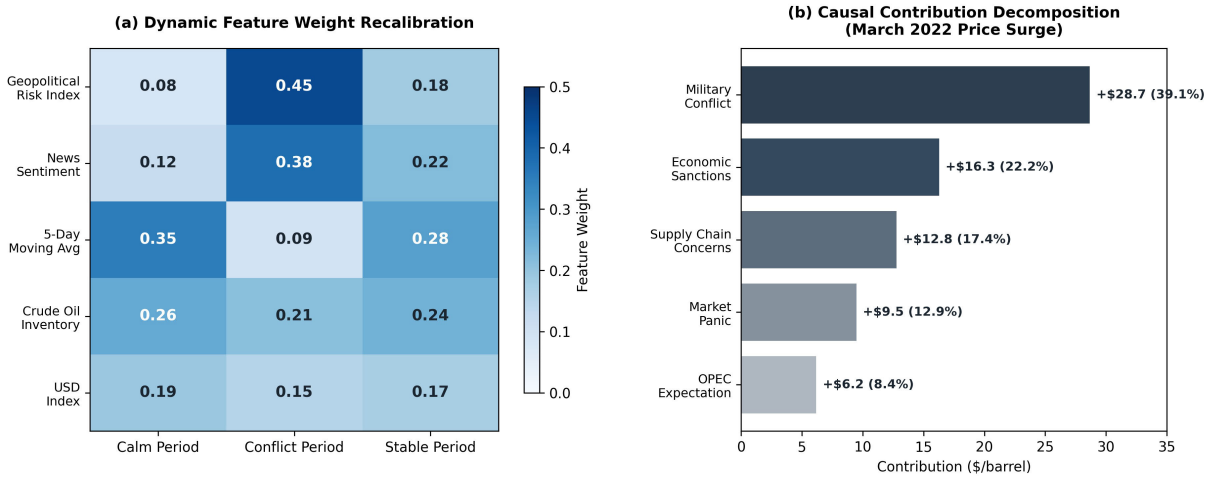


FIGURE 5. Dynamic feature analysis: feature weight recalibration across market states showing shifts in geopolitical risk index from 0.08 to 0.45 during conflict periods, and causal contribution decomposition of the March 2022 oil price surge with military conflict at 39.1% and economic sanctions at 22.2% as dominant drivers.

TABLE 3. Internal ablation study of the four-layer semantic extraction architecture.

Model Variant	RMSE	MAE	R^2	RMSE Change
Full model with four layers	5.67	4.38	0.903	—
Remove entity stance layer	5.91	4.56	0.894	+4.2%
Remove event type layer	5.84	4.51	0.897	+3.0%
Remove causal chain layer	5.99	4.64	0.889	+5.6%
Remove sentiment indicator layer	5.79	4.48	0.899	+2.1%
Retain sentiment indicator layer only	6.41	4.98	0.878	+13.1%

Note: RMSE Change is the percentage increase relative to the full model; positive values indicate performance degradation.

periods, prices are primarily driven by supply-demand fundamentals and technical indicators have strong predictive power, with our method maintaining a 7.7% advantage. During the event-intensive period, all model errors increase substantially, but our method shows the most significant advantage. LSTM error rises to \$18.45 with notably decreased prediction capability, while our method achieves \$8.92 error, a 34.7% improvement over TFT, validating the effectiveness of the proposed method in complex event-driven scenarios. During the stable period, our method achieves \$5.67 error with 10.6% improvement over TFT, an improvement magnitude between baseline and event-intensive periods.

From a prediction stability perspective, the error ratio between event-intensive and baseline periods is 3.53 times for LSTM but only 2.12 times for our method, indicating that the proposed improvements can effectively reduce the impact of extreme market environments on prediction performance. This characteristic has important practical value for risk management.

E. DYNAMIC FEATURE SELECTION EFFECT ANALYSIS

Table 5 presents the weight allocation patterns of major features across three typical periods, revealing how the model recalibrates prediction strategies according to environmental states.

During calm periods, technical indicators dominate, with the 5-day moving average ranking first at 0.35 weight while the Geopolitical Risk Index is marginal at only 0.08. During event-intensive periods, weight structure undergoes significant restructuring: the Geopolitical Risk Index rises from 0.08 to 0.45, hierarchical news sentiment rises to 0.38, while technical indicator weights drop substantially with the 5-day moving average falling to 0.09. This reflects how jump-like price movements break the continuity of technical patterns. During stable periods, weights rebalance: the Geopolitical Risk Index falls to 0.18 but remains higher than in calm periods, and technical indicator weights partially recover but do not fully return to calm period levels, reflecting that while markets have adapted to the new landscape, vigilance toward related risks is maintained.

TABLE 4. Prediction performance comparison across different periods.

Model	Baseline	Event-intensive	Stable	Overall
LSTM	5.23	18.45	7.89	8.76
BERT-Sent+LSTM	4.87	15.32	6.98	7.54
TFT without news	4.56	13.67	6.34	6.89
Ours	**4.21**	**8.92**	**5.67**	**5.67**
Ours vs. TFT improvement	7.7%	34.7%	10.6%	17.7%

Note: All errors are RMSE in USD per barrel.

TABLE 5. Feature weight distribution across different market states.

Feature	Calm Period	Event-intensive	Stable Period
Geopolitical Risk Index	0.08	**0.45**	0.18
Hierarchical News Sentiment	0.12	**0.38**	0.22
5-day Moving Average	**0.35**	0.09	**0.28**
Crude Oil Inventory	0.26	0.21	0.24
USD Index	0.19	0.15	0.17

Note: Weights represent relative contribution of each feature to prediction; column weights sum to 1; market state identification accuracy is 91.3%.

Static models use average weights of approximately 0.24, underestimating the importance of relevant risk indicators during event-intensive periods while over-responding to noise during calm periods. Dynamic models complete state determination and weight switching within a short time by analyzing daily text, while numerical statistical methods require several days of accumulation to achieve statistical significance. This response speed difference is particularly critical during rapid market transitions.

F. INTERPRETABILITY ANALYSIS AND COUNTERFACTUAL VALIDATION

Table 6 presents the factor contribution decomposition for the oil price increase from $54.3 to $127.8 in March 2022. The total increase of $73.5 is decomposed into five driving factors to validate the model's interpretability capabilities.

Military conflict contributes $28.7 at 39.1%, representing the largest single driving force. Counterfactual analysis removing all military action nodes predicts oil prices of approximately $99.1. Economic sanctions contribute $16.3 at 22.2%; the counterfactual scenario retaining military conflict but removing sanction edges predicts oil prices of $110.9, with sanction effects producing nonlinear amplification through causal chains in the graph structure. Supply chain concerns contribute $12.8 at 17.4%, market panic sentiment contributes $9.5 at 12.9%, and production cut expectations contribute $6.2 at 8.4%.

The total contribution of five factors at $73.5 matches the actual increase, satisfying additivity constraints. These factors are interconnected through causal chains: military conflict triggers sanctions, sanctions exacerbate supply concerns, and supply concerns stimulate market panic. Graph structure can capture this cascading propagation pattern. Factor contribution decomposition provides quantitative evidence for understanding price formation mechanisms, demonstrating the application value of interpretability analysis methods in event-driven prediction.

V. DISCUSSION

Experimental results validate the effectiveness of the three proposed improvement measures. The overall performance

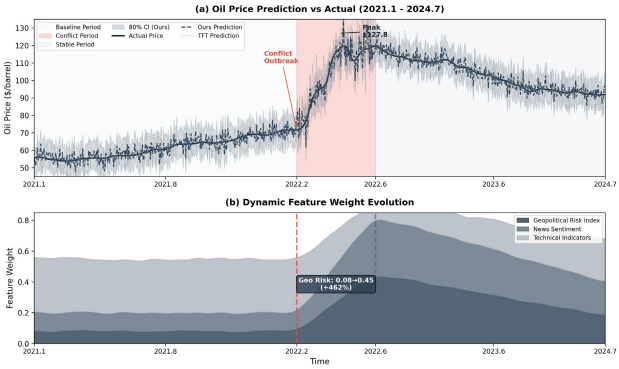


FIGURE 6. Temporal dynamics of prediction and adaptation: (a) oil price prediction versus actual values (2021.1–2024.7) with 80% confidence intervals across baseline, conflict, and stable periods, and (b) dynamic feature weight evolution demonstrating the 462% surge in geopolitical risk index weight during conflict outbreak.

improvement of 17.7% demonstrates that targeted improvements on existing multimodal fusion frameworks can yield substantial prediction accuracy gains. The 34.7% improvement during event-intensive periods indicates that hierarchical semantic extraction and dynamic feature selection show more pronounced advantages in scenarios with rich textual information and rapidly changing markets. The improvement in prediction stability also warrants attention: the error ratio between event-intensive and baseline periods decreased from 3.53 times to 2.12 times, indicating that the proposed method maintains relatively robust performance across different market states. Ablation experiments reveal independent contributions and synergistic mechanisms among modules. Hierarchical semantic extraction's single-step improvement of 6.4% validates the importance of fully preserving fine-grained textual information. Heterogeneous graph modeling and dynamic feature selection contribute 3.8% and 3.7% respectively. The three components form a complete chain from information extraction to relationship modeling to flexible utilization, with synergistic effects enabling overall performance improvement to exceed the simple sum of individual module contributions.

The effectiveness of hierarchical semantic extraction stems from thorough preservation of textual information.

TABLE 6. Factor contribution decomposition for March 2022 oil price increase.

Driving Factor	Contribution	Share	Counterfactual Scenario
Military Conflict	+28.7	39.1%	Remove military action nodes
Economic Sanctions	+16.3	22.2%	Remove sanction relation edges
Supply Chain Concerns	+12.8	17.4%	Assume normal supply
Market Panic Sentiment	+9.5	12.9%	Reduce fear index
Production Cut Expectations	+6.2	8.4%	Assume maintained production
Total	+73.5	100%	-

Note: Units are USD per barrel; counterfactual scenarios are implemented by removing corresponding nodes or edges from the heterogeneous graph and regenerating predictions.

TABLE 7. Temporal distribution statistics of news data.

Period	Time Range	Reports	Daily Avg.	Share
Pre-conflict	2021.1–2022.2	876	3.1	19.5%
Conflict outbreak	2022.3–2022.6	1,028	12.4	22.9%
Conflict continuation	2022.7–2022.12	612	4.8	13.6%
Validation	2023.1–2023.6	589	4.6	13.1%
Test	2023.7–2024.7	1,382	5.3	30.9%

TABLE 8. Entity type distribution.

Entity Type	Count	Examples
State actors	21	Russia, Saudi Arabia, USA
International organizations	12	OPEC, EU, NATO
Energy companies and institutions	150	Gazprom, Shell, IEA

TABLE 9. Event type distribution.

Event Type	Count	Share
Economic sanctions	1,059	23.6%
Supply changes	991	22.1%
Military actions	839	18.7%
Diplomatic negotiations	727	16.2%
Other	871	19.4%

Traditional flattened methods inevitably lose structured information when compressing complex text into single scalars, while hierarchical representation explicitly separates and encodes entity differences, event types, and causal relationships through a four-layer architecture. The pre-trained knowledge of large language models enables reasonable parsing even for event types with sparse samples in training data. The effectiveness of heterogeneous graph modeling stems from explicit characterization of network effects among events. Graph structure encodes complex interaction patterns such as cumulative effects, propagation effects, and synergistic effects through topological relationships of nodes and edges. Relation-aware attention mechanisms enable the model to dynamically adjust the importance of different relation types according to environmental states. The effectiveness of dynamic feature selection stems from directly addressing non-stationarity. The globally optimal feature set selected by static methods is essentially a compromise across all states, while dynamic methods decompose global optimization into state-specific local optimization through state identification. Semantic-driven state identification bypasses the lag of numerical statistical methods, enabling faster model response to environmental changes.

Although this paper uses oil price prediction as a validation case, the three proposed improvement measures possess good generality at the methodological level. Hierarchical semantic extraction methods are applicable to any prediction scenario requiring extraction of structured features from unstructured text, applicable to epidemic spread prediction, supply chain risk warning, financial market analysis, and other tasks. Different application scenarios require designing corresponding semantic hierarchy structures according to domain characteristics, but the overall architecture can be reused. Heterogeneous graph modeling methods are applicable to any prediction task where complex dependencies exist among entities and event impacts propagate through networks. Financial risk contagion, social media sentiment propagation, supply chain disruption prediction, and other scenarios can all draw upon the heterogeneous graph modeling approach. Dynamic feature selection methods are applicable to any non-stationary prediction environment with state switching. Economic cycles, epidemic propagation, market sentiment, and other domains all exhibit state-dependent characteristics of feature importance.

Current methods have several limitations requiring future research improvements. Regarding computational efficiency, reliance on large language model APIs for semantic extraction poses cost and latency constraints. Future work may explore domain-specific lightweight language models or hybrid architectures to reduce computational overhead. Regarding graph construction, automatically extracted causal relationships may contain noise, and structural dependency edges have low update frequency. Future work may develop continuous learning mechanisms to enable dynamic graph evolution. Regarding state modeling, discrete state space partitioning may lose information about gradual transitions between states. Future work may explore continuous state spaces or soft state assignment methods. Regarding validation scope, current experiments are conducted on a single asset only, and method effectiveness in other assets and domains awaits further verification.

VI. CONCLUSION

Event-driven time series forecasting tasks face challenges including insufficient utilization of textual semantic information, difficulty modeling network effects among events, and dynamic changes in feature importance under non-stationary environments. Existing methods have room for

TABLE 10. Auxiliary economic variable sources.

Variable	Data Source	Original Freq.	Alignment Method
Geopolitical Risk Index	Caldara and Iacoviello	Daily	Direct trading day match
Economic Policy Uncertainty Index	Baker, Bloom and Davis; FRED: USEPUINDXD	Daily	Direct trading day match
U.S. Dollar Index	FRED	Daily	Direct trading day match
S&P 500 Index	FRED	Daily	Direct trading day match
Crude Oil Inventory	EIA Weekly Petroleum Status Report	Weekly	Forward fill
OPEC Production	OPEC Monthly Oil Market Report	Monthly	Forward fill

improvement in these aspects. This study proposes three improvement measures building upon multimodal graph neural network and Temporal Fusion Transformer frameworks: LLM-based hierarchical semantic extraction decomposes text into four dimensions of entity stances, event types, causal chains, and sentiment indicators, preserving fine-grained information lost by flattened methods; heterogeneous information graph neural propagation modeling characterizes cumulative, propagation, and synergistic effects among events through graph structure; semantic-driven dynamic feature selection adaptively adjusts feature weights according to environmental states, responding quickly during state transitions.

Using oil price prediction as a validation case, experimental results show that the proposed method improves 17.7% over the Temporal Fusion Transformer baseline, with improvement reaching 34.7% during event-intensive periods and significantly enhanced prediction stability. Ablation experiments validate independent contributions of each improvement measure: hierarchical semantic extraction contributes 6.4%, heterogeneous graph modeling contributes 3.8%, and dynamic feature selection contributes 3.7%. Dynamic feature weight analysis and factor contribution decomposition demonstrate the model's interpretability capabilities.

This study provides an effective approach for multimodal information fusion in event-driven time series forecasting. The three proposed improvement measures possess generality at the methodological level and can provide reference for related tasks such as epidemic spread prediction, supply chain risk warning, and financial market analysis. Future research may further extend in areas including computational efficiency optimization, dynamic graph construction, continuous state modeling, and cross-domain validation.

AUTHOR CONTRIBUTIONS STATEMENT

Shang Cao conceived the study, designed the tripartite forecasting framework utilizing Large Language Models, and wrote the original draft of the manuscript. Ziqi Qu performed the data collection, implemented the baseline models, and prepared the visualizations (Figures 1–3). Caichun Cen acquired the funding support, provided essential computing resources for the experiments, and participated in the validation of the results. Zeling Xu and Yuntao Zou supervised the project, refined the causal inference methodology, and significantly revised the manuscript. All authors reviewed and approved the final manuscript.

APPENDIX

DATASET SUPPLEMENTARY DESCRIPTION

News data were collected from Reuters and Oilprice.com, then processed through deduplication and non-English text filtering, yielding 4,487 valid reports. The average input length per report after truncation is 417 words. The temporal distribution of news is uneven, with statistics for each period shown in Table 7.

A total of 183 entities were identified after extraction by the large language model, with their distribution shown in Table 8.

The distribution of event types according to the ontology classification is shown in Table 9.

The sources and frequencies of auxiliary economic variables are shown in Table 10.

News published on non-trading days is assigned to the next trading day. Weekly and monthly data are aligned to trading day frequency through forward filling, whereby the most recent observation is carried forward until new data are released.

REFERENCES

[1] D. Caldara and M. Iacoviello, "Measuring geopolitical risk," *Amer. Econ. Rev.*, vol. 112, no. 4, pp. 1194–1225, 2022.

[2] Y. Cai, V. Mignon, and J. Saadaoui, "Not all political relation shocks are alike: Assessing the impacts of US–China tensions on the oil market," *Energy Econ.*, vol. 114, 2022, Art. no. 106199.

[3] B. Lim and S. Zohren, "Time-series forecasting with deep learning: A survey," *Phil. Trans. Roy. Soc. A, Math., Phys. Eng. Sci.*, vol. 379, no. 2194, 2021, Art. no. 20200209.

[4] K. Benidis, S. S. Rangapuram, V. Flunkert, Y. Wang, D. Maddix, C. Turkmen, J. Gasthaus, M. Bohlke-Schneider, D. Salinas, L. Stella, F.-X. Aubet, L. Callot, and T. Januschowski, "Deep learning for time series forecasting: Tutorial and literature survey," *ACM Comput. Surv.*, vol. 55, no. 6, pp. 1–36, Jul. 2023.

[5] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, pp. 11106–11115.

[6] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Proc. 35th Int. Conf. Neural Inf. Process. Systems (NIPS)*. Red Hook, NY, USA: Curran Associates Inc., 2021, pp. 22419–22430, Paper 1717.

[7] H. Wu, H. Zhou, M. Long, and J. Wang, "Interpretable weather forecasting for worldwide stations with a unified deep model," *Nature Mach. Intell.*, vol. 5, no. 6, pp. 602–611, Jun. 2023.

[8] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirnsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, G. Holland, O. Vinyals, J. Stott, A. Pritzel, S. Mohamed, and P. Battaglia, "Learning skillful medium-range global weather forecasting," *Science*, vol. 382, no. 6677, pp. 1416–1421, Dec. 2023.

[9] A. H. Huang, H. Wang, and Y. Yang, "FinBERT: A large language model for extracting information from financial text," *Contemp. Accounting Res.*, vol. 40, no. 2, pp. 806–841, May 2023.

[10] A. Lopez-Lira and Y. Tang, "Can ChatGPT forecast stock price movements? Return predictability and large language models," 2023, *arXiv:2304.07619*.
[11] A. Lopez-Lira and Y. Tang, "Can chatgpt forecast stock price movements? return predictability and large language models (2023)," 2023, *arXiv:2304.07619*.
[12] I. Price, A. Sanchez-Gonzalez, F. Alet, T. R. Andersson, A. El-Kadi, D. Masters, T. Ewalds, J. Stott, S. Mohamed, P. Battaglia, R. Lam, and M. Willson, "Probabilistic weather forecasting with machine learning," *Nature*, vol. 637, no. 8044, pp. 84–90, 2025.
[13] X. Fu, J. Zhang, Z. Meng, and I. King, "MAGNN: Metapath aggregated graph neural network for heterogeneous graph embedding," in *Proc. Web Conf.*, 2020, pp. 2331–2341.
[14] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *Int. J. Forecasting*, vol. 37, no. 4, pp. 1748–1764, Oct. 2021.
[15] T. Zhou, Z. Ma, X. Wang, L. Sun, R. Jin, and Q. Wen, "Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting," in *Proc. Int. Conf. Mach. Learn.*, 2026, pp. 10–34.
[16] T. Loughran and B. McDonald, "Textual analysis in finance," *Annu. Rev. Financial Econ.*, vol. 12, no. 1, pp. 357–375, 2020.
[17] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol.*, vol. 1, 2019, pp. 4171–4186.
[18] S. Ozturk, M. B. Yilmaz, M. Kara, M. Talat Yavuz, A. Koç, and T. Çukur, "Meta-entity driven triplet mining for aligning medical vision-language models," 2025, *arXiv:2504.15929*.
[19] F. Xing, K. Du, G. Mengaldo, E. Cambria, and R. Welsch, "AI reshaping financial modeling," *npj Artif. Intell.*, vol. 1, no. 1, p. 29, Oct. 2025.
[20] Ş. Öztürk, A. Güngör, and T. Çukur, "Diffusion probabilistic models for image formation in mri," in *Generative Machine Learning Models in Medical Image Computing*. Cham, Switzerland: Springer, 2024, pp. 341–360.
[21] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," 2022, *arXiv:2211.14730*.
[22] D. Xu, W. Chen, W. Peng, C. Zhang, T. Xu, X. Zhao, X. Wu, Y. Zheng, Y. Wang, and E. Chen, "Large language models for generative information extraction: A survey," *Frontiers Comput. Sci.*, vol. 18, no. 6, Dec. 2024, Art. no. 186357.
[23] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3580–3599, Jul. 2024.
[24] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.
[25] B. Jiang, Z. Zhang, D. Lin, J. Tang, and B. Luo, "Semi-supervised learning with graph learning-convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 11313–11320.
[26] V. Petar, C. Guillem, C. Arantxa, R. Adriana, L. Pietro, and B. Yoshua, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, vol. 8, 2018.
[27] S. Yildiz, A. MEMiş, and S. Varli, "TRCaptionNet++: A high-performance encoder–decoder based deep Turkish image captioning model fine-tuned with a large-scale set of pretrain data," *Turkish J. Electr. Eng. Comput. Sci.*, vol. 33, no. 5, pp. 669–687, Sep. 2025.
[28] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, and P. S. Yu, "Heterogeneous graph attention network," in *Proc. World Wide Web Conf.*, 2019, pp. 2022–2032.
[29] E. Budur and T. Güngör, "Conversational open-domain question answering for resource-constrained languages," *Turkish J. Electr. Eng. Comput. Sci.*, vol. 33, no. 2, pp. 203–223, Mar. 2025.
[30] S. Xiang, D. Cheng, C. Shang, Y. Zhang, and Y. Liang, "Temporal and heterogeneous graph neural network for financial time series prediction," in *Proc. 31st ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2022, pp. 3584–3593.
[31] M. Segnon, R. Gupta, and B. Wilfling, "Forecasting stock market volatility with regime-switching GARCH-MIDAS: The role of geopolitical risks," *Int. J. Forecasting*, vol. 40, no. 1, pp. 29–43, Jan. 2024.
[32] Y. Wang, A. Smola, D. C. Maddix, J. Gasthaus, D. P. Foster, and T. Januschowski, "Deep factors for forecasting," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6607–6617.
[33] Y. Zou, Z. Lin, Q. Zhang, Z. Liu, and Z. Xu, "Exploration-guided adaptive learning from mixed-quality data for satellite multi-battery management," *J. Energy Storage*, vol. 150, Mar. 2026, Art. no. 120546.
[34] Y. Zou, Z. Lin, Q. Zhang, Z. Liu, and Z. Xu, "Large language models enable semantic-guided hierarchical games for intelligent battery coordination," *Adv. Eng. Informat.*, vol. 71, Apr. 2026, Art. no. 104312.
[35] S. Gunasekaran, A. Kembay, H. Ladret, R.-J. Zhu, L. Perrinet, O. Kavehei, and J. Eshraghian, "A predictive approach to enhance time-series forecasting," *Nature Commun.*, vol. 16, no. 1, p. 8645, Sep. 2025.

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